Physics A (H156, H556)

Unified - Electric fields ppq
KSA Physics

Please note that you may see slight differences between this paper and the original.

Candidates answer on the Question paper.

OCR supplied materials:
Additional resources may be supplied with this paper.

Other materials required:
- Pencil
- Ruler (cm/mm)

Duration: Not set

INSTRUCTIONS TO CANDIDATES

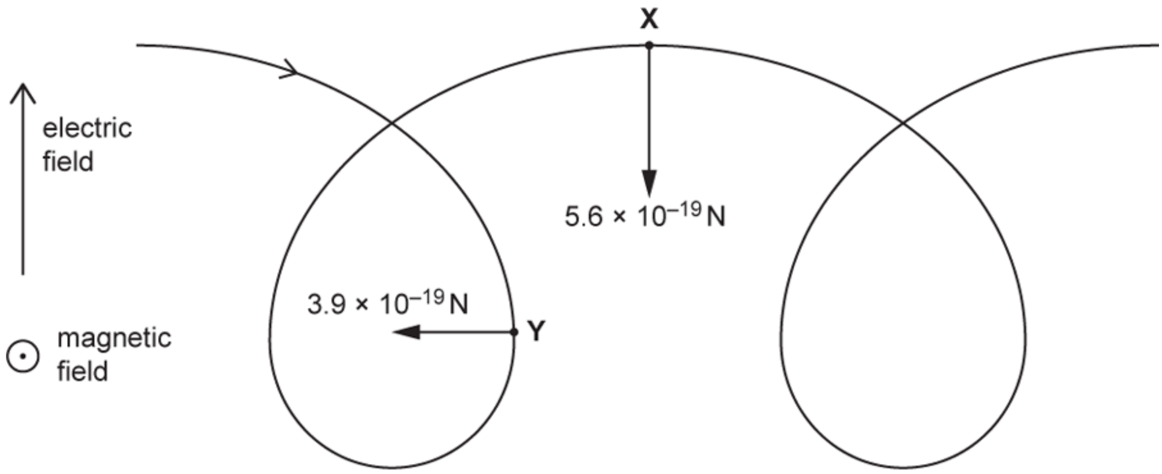
- Write your name, centre number and candidate number in the boxes above. Please write clearly and in capital letters.
- Use black ink. HB pencil may be used for graphs and diagrams only.
- Answer **all** the questions, unless your teacher tells you otherwise.
- Read each question carefully. Make sure you know what you have to do before starting your answer.
- Where space is provided below the question, please write your answer there.
- You may use additional paper, or a specific Answer sheet if one is provided, but you must clearly show your candidate number, centre number and question number(s).

INFORMATION FOR CANDIDATES

- The quality of written communication is assessed in questions marked with either a pencil or an asterisk. In History and Geography a *Quality of extended response* question is marked with an asterisk, while a pencil is used for questions in which *Spelling, punctuation and grammar and the use of specialist terminology* is assessed.
- The number of marks is given in brackets **[ ]** at the end of each question or part question.
- The total number of marks for this paper is **49**.
- The total number of marks may take into account some 'either/or' question choices.

1(a) The figure below shows the path of a proton moving in a region occupied by both an electric field and a magnetic field.

The direction of the electric field lines is perpendicular to the direction of the magnetic field lines.



The uniform electric field is directed upwards, with electric field strength E = 0.90 N C^{-1}.

The uniform magnetic field is directed out of the plane of the paper, with magnetic flux density $B = 5.0 \times 10^{-5}$ T.

At point **X** the proton is moving horizontally to the right. The magnitude of the **magnetic** force at **X** is 5.6×10^{-19} N.

At point **Y** the proton is moving vertically downwards. The magnitude of the **magnetic** force at **Y** is 3.9×10^{-19} N.

The **electric** forces acting on the proton at **X** and **Y** are **not** shown in the figure.

Show that the magnitude of the constant **electric** force acting on the proton is about 10^{-19} N.

[1]

(b)

i. Suggest why the **magnetic** force acting on the proton has a different magnitude at X than at Y.

-- **[1]**

ii. At X, the motion of the proton is instantaneously equivalent to motion in a circle at a constant speed.

Calculate the radius of this circular motion.

radius = ... m **[4]**

iii. **1** Calculate the magnitude of the resultant force on the proton at Y.

resultant force = ... N **[2]**

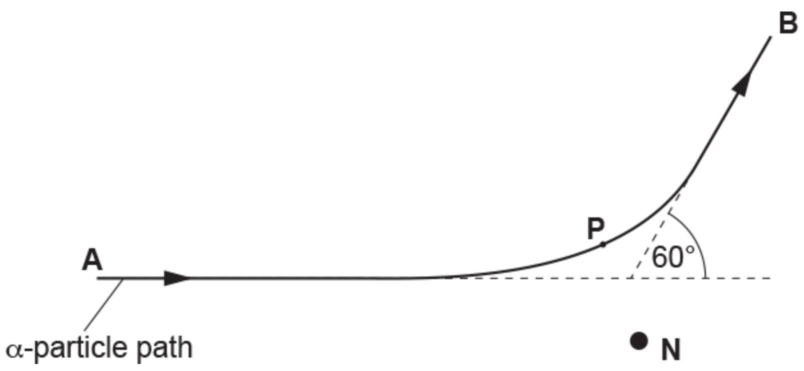
2 Explain why the motion of the proton at Y is **not** instantaneously equivalent to motion in a circle at a constant speed.

-- **[2]**

2(a) A beam of α-particles is incident on a thin gold foil. Most α-particles pass straight through the foil. A few are deflected by gold nuclei.

The diagram shows the path of one α-particle which passes close to a gold nucleus **N** in the foil. The α-particle is deflected through an angle of 60° as it travels from **A** to **B**.

P marks its position of closest approach to the gold nucleus.



Another α-particle in the beam is deflected by the same gold nucleus **N** through an angle of 30°.

Sketch its path onto the diagram above. **[2]**

(b) The distance between **P** and **N** is 6.8×10^{-14} m.

Calculate the magnitude of the electrostatic force *F* between the α-particle $({}^{4}_{2}He)$ and the gold nucleus $({}^{197}_{79}Au)$

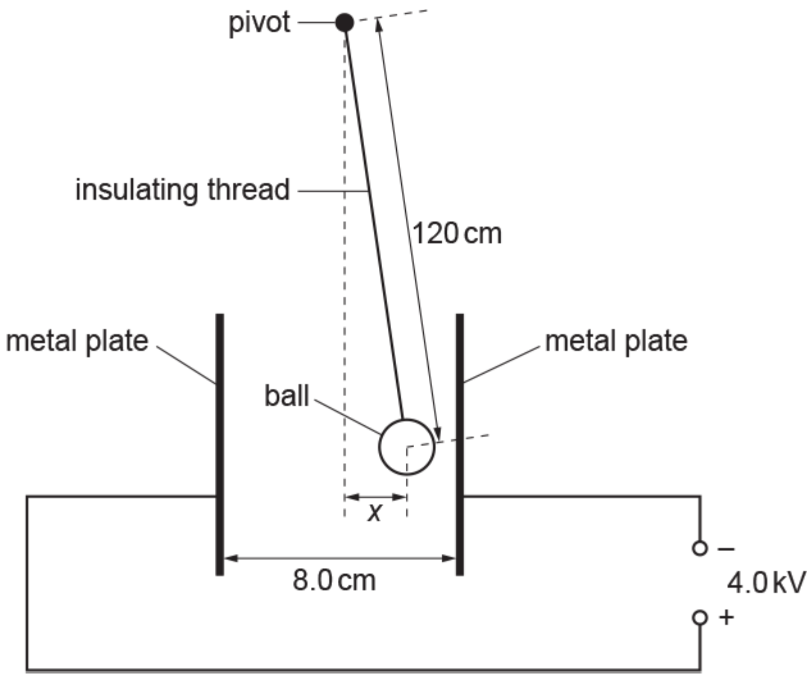
when the α-particle is at **P**.

F = ... N **[4]**

3(a) A ball coated with conducting paint has weight 0.030 N and radius 1.0 cm. The ball is suspended from an insulating thread. The distance between the pivot and the centre of the ball is 120 cm.

The ball is placed between two vertical metal plates. The separation between the plates is 8.0 cm. The plates are connected to a 4.0 kV power supply.

The ball receives a positive charge of 9.0 nC when it is made to touch the positive plate. It then repels from the positive plate and hangs in equilibrium at a displacement *x* from the vertical, as shown below. The diagram is **not** drawn to scale.



i. Show that the electric force acting on the charged ball is 4.5×10^{-4} N.

[2]

ii. Draw, on the diagram above, arrows which represent the **three** forces acting on the ball. Label each arrow with the name of the force it represents.

[2]

iii. By taking moments about the pivot, or otherwise, show that x = 1.8 cm.

[2]

(b) The ball is still positively charged.

The plates are now moved slowly towards each other whilst still connected to the 4.0 kV power supply. The plates are stopped when the separation is 5.0 cm.

Explain the effect that this has on the deflection of the ball and explain why the ball eventually starts to oscillate between the plates.

[4]

(c) When the ball oscillates between the plates, the current in the external circuit is 3.2×10^{-8} A.

A charge of 9.0 nC moves across the gap between the plates each time the ball makes one complete oscillation.

Calculate the frequency *f* of the oscillations of the ball.

f = ... Hz **[2]**

4 A small thin rectangular slice of semiconducting material has width a and thickness b and carries a current I.
The current is due to the movement of electrons. Each electron has charge $-e$ and mean drift velocity v.
A uniform magnetic field of flux density B is perpendicular to the direction of the current and the top face of the slice as shown in Fig. 2.1.

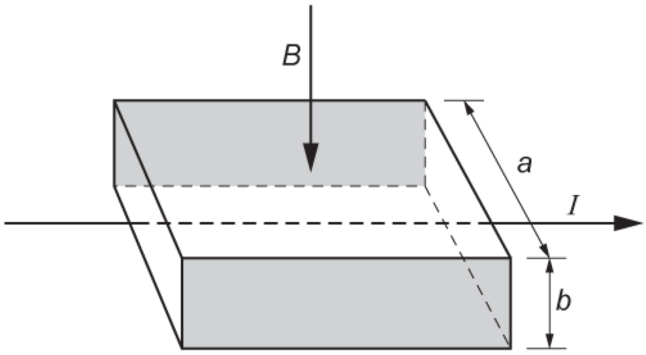


Fig. 2.1

As soon as the current is switched on, the moving electrons in the current are forced towards the shaded rear face of the slice where they are stored. This causes the shaded faces to act like charged parallel plates. Each electron in the current now experiences both electric and magnetic forces. The resultant force on each electron is now zero.

Write the expressions for the electric and magnetic forces acting on each electron and use these to show that the magnitude of the potential difference V between the shaded faces is given by

$$V = Bva.$$

[3]

5

Fig. 3.1 shows a simple representation of a hydrogen iodide molecule. It consists of two ions $^{1}_{1}H^{+}$ and $^{127}_{53}I^{-}$, held together by electric forces.

Fig. 3.1

i. Draw on Fig. 3.1 a minimum of five lines to show the electric field pattern between the ions.

[2]

ii. The charge on each ion has a magnitude *e* of 1.6×10^{-19} C. The ions are to be treated as point charges 5.0 $\times 10^{-10}$ m apart. Calculate the magnitude of the resultant electric field strength *E* at the **mid-point** between the ions.

E = ______________________________ N C^{-1} **[4]**

6(a) Electrons in a beam are accelerated from rest by a potential difference V between two vertical plates before entering a uniform electric field of electric field strength E between two horizontal parallel plates, a distance $2d$ apart.

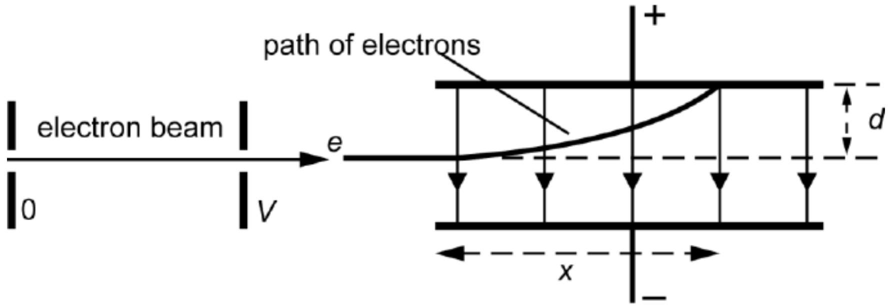


Fig. 2.1

The path of the electrons is shown in Fig. 2.1. The electron beam travels a horizontal distance x parallel to the plates before hitting the top plate. The beam has been deflected through a vertical distance d.

Show that x is related to V by the equation

$$x^2 = \frac{4dV}{E}.$$

[5]

(b) For different values of the accelerating p.d. V, the horizontal distance x is recorded. A table of results is shown with a third column giving values of x^2 including the absolute uncertainties.

V / V	x / cm	x^2 / cm^2
500	3.3 ± 0.1	10.9 ± 0.7
600	3.6 ± 0.1	13.0 ± 0.7
700	3.9 ± 0.1	15.2 ± 0.8
800	4.2 ± 0.1	17.6 ± 0.8
900	4.5 ± 0.1	20.3 ± 0.9
1000	4.7 ± 0.1	

i. Complete the missing value in the table, including the absolute uncertainty.

[1]

ii. Fig. 2.2 shows the axes for a graph of x^2 on the *y*-axis against *V* on the *x*-axis. The first four points have been plotted including error bars for x^2. Use data from the table to complete the graph.

[2]

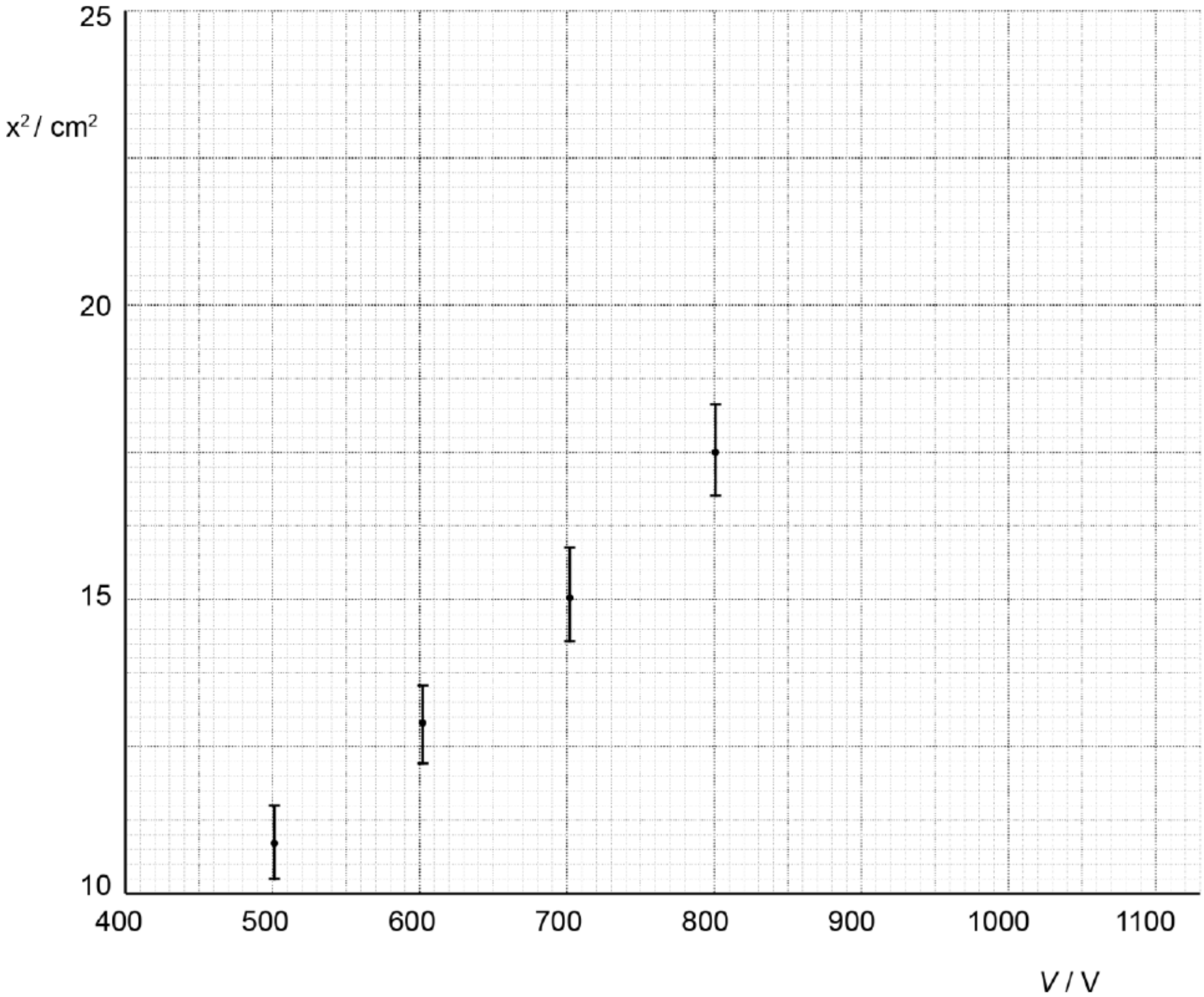


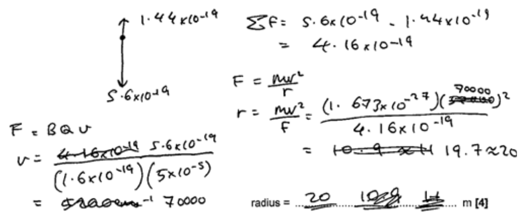
Fig. 2.2

iii. The separation of the horizontal plates is 4.0 ± 0.1 cm.
Use the graph to determine a value for E. Include the absolute uncertainty and an appropriate unit in your answer

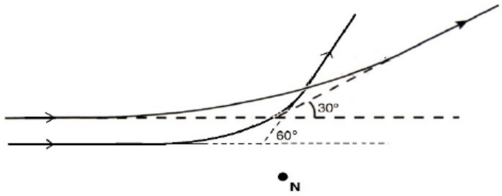
E = ______________ ±___________ unit___________ **[4]**

END OF QUESTION PAPER

Question			Answer/Indicative content	Marks	Guidance
1	a		F (= EQ) = 0.90 × 1.60 × 10^{-19} = 1.4(4) × 10^{-19} (N)	B1	Working and answer must both be shown Answer must be given to 2sf or more Unit need not be given but, if given, must be correct <u>Examiner's Comments</u> This was an easy introduction to the question, which used the definition of electric field strength; $E = F_E / Q$. Being a 'show that' question, candidates needed to show their working in full, including writing the value for the electronic charge (rather than simply 'e') and giving the answer to at least 2 s.f.
	b	i	($F = BQv$ but B and Q are constant, so) (the magnitude of) the velocity is different /changes	B1	**Allow** speed **Ignore** the direction is different <u>Examiner's Comments</u> The force on a charged particle moving at right angles to a magnetic field is given by the formula $F_{mag} = BQv$. Since B and Q are constants in this case, the reason for the different magnitude of F must be that the proton has a different velocity, v. **Common problems in 6(b)(i)** • using the formula $F = BIl\sin\theta$ and suggesting that the proton might be travelling at a different angle to the field, not realising that the proton is always travelling at right angles to the magnetic field in this question • suggesting that the proton may be in a weaker (or stronger) field at X than at Y, not realising that the magnetic field is uniform and so its field strength is constant throughout
		ii	$v = (\frac{F_{mag}}{BQ} =) \frac{5.6 \times 10^{-19}}{5.0 \times 10^{-5} \times 1.60 \times 10^{-19}}$ resultant force F_R= (5.6 – 1.4) × 10^{-19} $r = \left(\frac{mv^2}{F_R} =\right) \frac{1.673 \times 10^{-27} \times (7.0 \times 10^4)^2}{4.2 \times 10^{-19}}$ r = 20 (m) Alternative all-in-one method:	C1 C1 C1 A1 (C1) (C1) (C1) (A1)	$v = 7.0 \times 10^4$ (m s^{-1}) implies first C1 **Allow** 10^{-19} for 1.4×10^{-19} (giving F_R= 4.6 × 10^{-19}) F_R = 4.2 × 10^{-19} implies second C1 Do not credit if used as F_{mag} in F_{mag} in $F_{mag} = BQv$ Third C1 is for correct substitution into formula **Allow** m_p = 1.67 × 10^{-27} kg given to 3 s.f. **Not** m_p = 1.661 × 10^{-27} kg or m_p = 1.675 × 10^{-27} kg **Allow ECF** for incorrect v

Question			Answer/Indicative content	Marks	Guidance
			$r = \dfrac{m{F_{mag}}^2}{F_R\, B^2 Q^2}$ resultant force F_R= (5.6 – 1.4) × 10^{-19} $r = \dfrac{1.673 \times 10^{-27} \times (5.6 \times 10^{-19})^2}{4.2 \times 10^{-19} \text{ x } (5.0 \times 10^{-5})^2 \times (1.60 \times 10^{-19})^2}$ r = 20 (m)		Use of F_R = 5.6 × 10^{-19} or = 1.4 × 10^{-19} is **XP** **Allow** r = 19 (m) F_R = **4.2 × 10^{-19}** (4.16 × 10^{-19} to 3sf) implies second C1 An incorrect value of F_R is **XP** from this point Third C1 is for correct substitution into formula **Allow** r = 19 (m) <u>Examiner's Comments</u> This question could not be done in one step, by equating the magnetic force to the centripetal force. This is because, at X, the centripetal force is being provided by a combination of forces from both the electric and the magnetic field. The easiest approach is to find the velocity of the proton using $F_{mag} = BQv$ (the value for F_{mag} is given in the diagram as 5.6 × 10^{19} N). This velocity v can then be used in the formula $F = mv^2/r$ in order to calculate the radius, r. F here is the *resultant* force towards the centre of the circle, which is found from magnetic force downwards - electric force upwards (the electric force having been calculated in part (a)). Exemplar 3 is an example of a correct answer, clearly written to show each stage in the calculation: Exemplar 3 

Question			Answer/Indicative content	Marks	Guidance
		iii	$\lvert\text{resultant force}\rvert = (\sqrt{3.9^2 + 1.4^2}) \times 10^{-19}$ |resultant force| = 4.1×10^{-19} (N)	C1 A1	**Ignore** attempt to calculate weight of proton **Allow** $F_E = 10^{-19}$ **Allow** $\lvert F\rvert = 4.0 \times 10^{-19}$ (N) using $F_E = 1.0 \times 10^{-19}$ **Allow** $\lvert F\rvert = 4.2 \times 10^{-19}$ (N) using $F_E = 1.44 \times 10^{-19}$ <u>Examiner's Comments</u> There are two forces acting on the proton at Y: an electric force upwards (given in (a)) and a magnetic force to the left (shown on the diagram). These two forces act at right angles to each other, and so the magnitude of their resultant can be found using Pythagoras's Theorem. Credit was given for using a value for the electric force to 1, 2 or more significant figures.
		iv	<u>resultant / net</u> force is not perpendicular to velocity work is done on proton (therefore kinetic energy changes so speed is not constant)	B1 B1	**Allow** direction of motion / path but **not** speed for velocity **Allow** acceleration / <u>resultant</u> force is not (always) towards centre (of circle) **Allow** electric force is not perpendicular to velocity / is in the same direction as velocity **Ignore** references to centripetal **Ignore** references to centripetal <u>Examiner's Comments</u> At Y, the proton is moving downwards, with a resultant force being the combination of an electric force upwards and a magnetic force to the left (calculated in part 1). The resultant force cannot be at right angles to the velocity, so we cannot have circular motion. The component of the resultant force acting in the direction of the proton's motion will do work on the proton and change its speed. So, the proton cannot be travelling at a constant speed.
			Total	10	

Question			Answer/Indicative content	Marks	Guidance
2	a			B1 B1	Path is initially horizontal **and** further up the page than original Path <u>ends</u> at 30^0 to horizontal (angle must be labelled) in the direction shown <u>Examiner's Comments</u> The common errors here were: • not realising that, for the particle to be deflected through a smaller angle, it needed to be travelling further away from N • not labelling the final angle of 30° • not adding arrows to show the direction of travel • drawing a path that continued bending beyond the stated 30° (usually ending up parallel to the original path).
	b		Q = 79e and q = 2e $F = (1/4\pi\varepsilon_0)Qq/r^2$ $= 79 \times 2 \times (1.6 \times 10^{-19})^2 / [4\pi \times 8.85 \times 10^{-12} \times (6.8 \times 10^{-14})^2]$ = 7.9 (N)	C1 C1 C1 A1	**Apply ECF** for wrong charge(s), e.g. Q and / or q = e, or Q = 79 and / or q = 2, etc <u>Examiner's Comments</u> The most common error here was to use incorrect values for the charges on the two ions. Even so, most candidates were able to gain most of the marks with ECF.
			Total	6	

Question			Answer/Indicative content	Marks	Guidance
3	a	i	$F = QE = QV / d$ **or** $E = 5(.0) \times 10^4$ (Vm^{-1}) $F = 9.0 \times 10^{-9} \times 4000 / 8.0 \times 10^{-2}$ (= 4.5×10^{-4} N)	C1 A1	$F = 5.0 \times 10^4 \times 9.0 \times 10^{-9}$ <u>Examiner's Comments</u> Many lower ability candidates did not appreciate the uniform nature of the electric field between the plates and attempted to use Coulomb's Law.
		ii	weight; arrow vertically downwards tension; arrow upwards in direction of string electric (force); arrow horizontally to the <u>right</u> (not along dotted line)	B1 x 2	All correct, 2 marks; 2 correct, 1 mark 1 mark maximum if more than 3 arrows are drawn **Ignore** position of arrows **Allow** W **or** 0.030(N) (**not** gravity or g) **Allow** T **Allow** F **or** E or 4.5×10^{-4}(N) or electrostatic **Ignore** repulsion or attraction **Not** electric field / electric field strength / electromagnetic <u>Examiner's Comments</u> Most candidates scored a mark for showing the weight and tension forces accurately. Only a small proportion labelled the electric force arrow correctly and drew it as clearly perpendicular to the plates. **AfL** Do not use the word 'gravity' in place of 'weight'
		iii	$W x = F l$ $0.03 x$ $= 4.5 \times 10^{-4} \times 120$ **or** $= 4.5 \times 10^{-4} \times 1.2$ $x = 1.8$ cm **or** $x = 0.018$ m	M1 M1 A0	**Allow** any valid alternative approach e.g. M1 deflection angle $\theta = 1°$ M1 x = 120sinθ 1 mark for each side of the equation <u>Examiner's Comments</u> Although most candidates knew the principle of moments, many were unable to apply it correctly in this situation. More practice at this sort of question is recommended.

Question			Answer/Indicative content	Marks	Guidance
	b		Electric force/field (strength) increases Ball deflected further from vertical / moves to the right / touches negative plate Ball acquires the charge of the (negative) plate when it touches (Oscillates because) constantly repelled from (oppositely) charged plate	B1 B1 B1 B1	Must be clear which force is increasing Must have the idea of a repeating cycle <u>Examiner's Comments</u> The purpose of this question was to challenge the candidates to use their knowledge of electric fields in a novel practical situation. The word 'oscillate' confused many candidates, who tried to explain why the ball would perform simple harmonic motion.
	c		$I = Qf$ or $Q = It$ $f = 3.2 \times 10^{-8}/9.0 \times 10^{-9} = 3.6$ (Hz)	C1 A1	
			Total	**12**	

Question			Answer/Indicative content	Marks	Guidance
4			$F = Bev$ **and** $F = eE$ $E = V / a$ **or** $F = (eE) = eV / a$ $Bev = eV / a$ giving $V = Bva$	B1 B1 B1	**allow** Q or q for e **Examiner's Comments** This was an exercise in writing basic definitions in algebraic form and then using them to derive a given equation. More than half of the candidates managed to gain full marks with less than one third scoring zero. The presentation was sometimes difficult to follow with the inclusion of unnecessary equations and deletions and the substitution of *d* for *a* in the last line.
			Total	3	
5		i	acceptable pattern with lines touching but not entering spheres	B1	adequate drawing for 1 mark
		i	lines perpendicular to spheres and arrows from plus ion to minus ion	B1	award second mark for detail/quality
		ii	$E = kQ/r^2$ where $k = 1/4\pi\varepsilon_0$	C1	correct formula with Q = e
		ii	$E = 9 \times 10^9 \times 1.6 \times 10^{-19}/6.25 \times 10^{-20}$	C1	correct substitution
		ii	$E = 2.3 \times 10^{10}$	C1	evaluation
		ii	$2E = 4.6 \times 10^{10}$ ($N\ C^{-1}$)	A1	fields of charges add, **allow ecf** for E
			Total	6	

Question			Answer/Indicative content	Marks	Guidance
6	a		$eV = ½mv^2$ so $v^2 = 2eV/m$ $ma = eE$ so $a = eE/m$ $x = vt$ $d = ½at^2 = ½a(x/v)^2$ $d = (eE/2m).x^2.(m/2eV) = Ex^2/4V$ $x^2 = 4(d/E)V$	B1 B1 B1 B1 B1 A0	four equations are needed and some sensible substitution, etc. shown for the fifth mark
	b	i	22.1 ± 0.9	B1	value plus uncertainty both required for the mark allow ± 1.0
		ii	two points plotted correctly, including error bars;	B1	**ecf** value and error bar of first point
		ii	line of best fit worst acceptable straight line.	B1	**allow ecf** from points plotted incorrectly steepest or shallowest possible line that passes through all the error bars; should pass from top of top error bar to bottom of bottom error bar **or** bottom of top error bar to top of bottom error bar
		ii	gradient (= 4d / E) = 2.4 ± 0.4;	B1	**allow** 2.4 ± 0.5
		ii	$E = 4 \times 2.0 \times 10^{-2} / 2.4 \times 10^{-6} = 3.3 \times 10^4$	B1	
		ii	$(3.3) \pm 0.6 \times 10^4$	B1	0.1/4 + 0.4/2.4 = 0.192 × 3.3 = 0.63
		ii	V m^{-1} or N C^{-1}	B1	0.1/4 + 0.5/2.4 = 0.233 × 3.3 = 0.77 **allow** $3.3 \pm 0.8 \times 10^4$
			Total	**12**	